Unbounded prime orders in the checked arithmetic family

For every bound B, Lean proves an actual parameter t with g=12t+2>B and prime n=24t+5>B. The reduced-pair arithmetic group over F_3 for $f=X^n+1$ has cardinality n, actual generator (X,1) of exact order n, certified normalized Cantor sums for every ordered pair, and every member generated by that point. [pcc:unverified]

All members are smooth, fraction 1. The full finite universe of valid reduced pairs has strictly smaller fraction. Such prime-order disparities exist beyond every bound; Lean directly negates universal equality for the stated prime-order arithmetic family. [pcc:unverified]

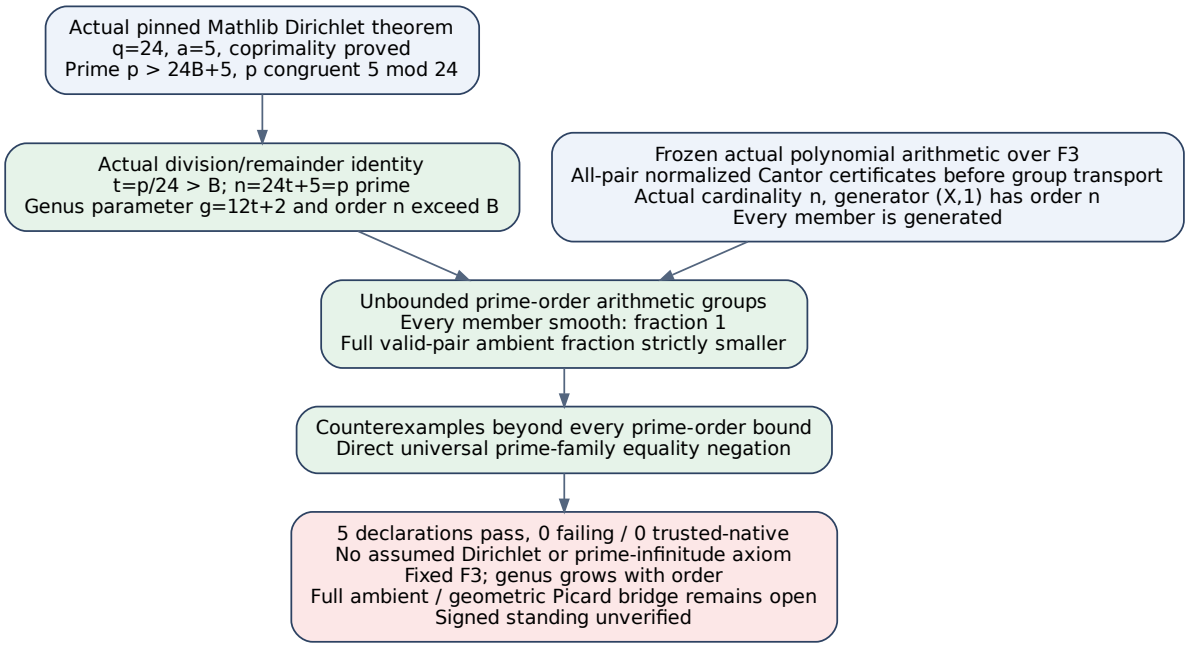

Figure 1: Actual Dirichlet theorem and arithmetic certificates give unbounded prime orders. [pcc:unverified]

Parameter proof, exact declarations and scope

Mathlib's actual Nat.forall_exists_prime_gt_and_modEq theorem applies to q=24, a=5, with nonzero modulus and coprimality discharged in the kernel. It gives a prime p>24B+5 with p≡5 mod 24. The actual modular and division/remainder identities give t=p/24>B and 24t+5=p. This is an application of the proved Dirichlet theorem, with its transitive axioms audited. No Dirichlet or prime-infinitude premise is assumed. [pcc:unverified]

The frozen parent proves actual polynomial outputs and closure before group transport, then proves every group sum equals those normalized outputs. Its actual cardinality/order/generation and smooth-fraction results apply at the chosen prime parameter. The uniform nonsplit pair (X^2+X+2,X) remains outside the family. No arithmetic or geometric map is introduced as an assumption. [pcc:unverified]

Declaration in MumfordPrimeFamilyAudit	Exact obligation
prime_exponent_unbounded	Every B has t>B with 24t+5 prime
prime_cardinality_and_order	Actual group cardinality prime, generator order exact, genus/order exceed B
unbounded_prime_family_certificate	Prime order, all normalized sums, generation and strict fraction inequality beyond every bound
unbounded_prime_smoothness_counterexamples	Existential prime-order disparities beyond every order bound
universal_prime_family_uniformity_false	Direct negation of universal equality on this prime-order arithmetic family

The full project gate passes five declarations, zero failures and zero trusted-native exceptions. Lean 4.35.0 / Mathlib 56a00585. Only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, linter waivers or added elaboration settings. All five attempt/source files and five gate files have byte-equal archive checks. Issued parent formal sources remain unchanged. Exact commit, packet, actual verdict and signed standing appear in verification.json. Signed standing remains unverified. [pcc:unverified]

The pinned Mathlib PrimesInAP.lean introduction, proof outline and used theorem statements/proofs were read; its author is Michael Stoll (2024 header). The exact formal source URL, hash and revision are retained. The parent Stoll 2014 and Jędrzejak 2016 read receipts and known torsion-family attribution remain unchanged. No new full-paper read or novelty claim. [pcc:unverified]

The base field is fixed at F_3 and the genus parameter grows with order. The full ambient reduced-pair group and geometric Picard correspondence remain separate formal obligations. No claim about a prime whole-Jacobian order, a typical/dominant-prime subgroup or fixed-genus $q\to\infty$ asymptotics is made. [pcc:unverified]

No native experiment, fresh sampling, timing or paid compute. KB MCP absent, wrapper 127; bounded primary formal-source/repository/Mathlib fallback. Conductor unavailable. Proof map and exact pins: README.md, sources.json and verification.json. [pcc:unverified]